



ParkBot Assist

Intelligent Parking, Powered by AI

Multi-Level Car Parking - Slot Discovery, Booking & Feedback Intelligence

IBM Ecosystem Engineering Labs



THE PROBLEM

Why parking remains a frustrating experience



Wasted Time

Drivers circle for 15-20 mins on average searching for an open slot in large MLCP facilities.



No Real-Time Info

Static signage cannot reflect live occupancy. Users have zero visibility before entering the facility.



No Time Booking

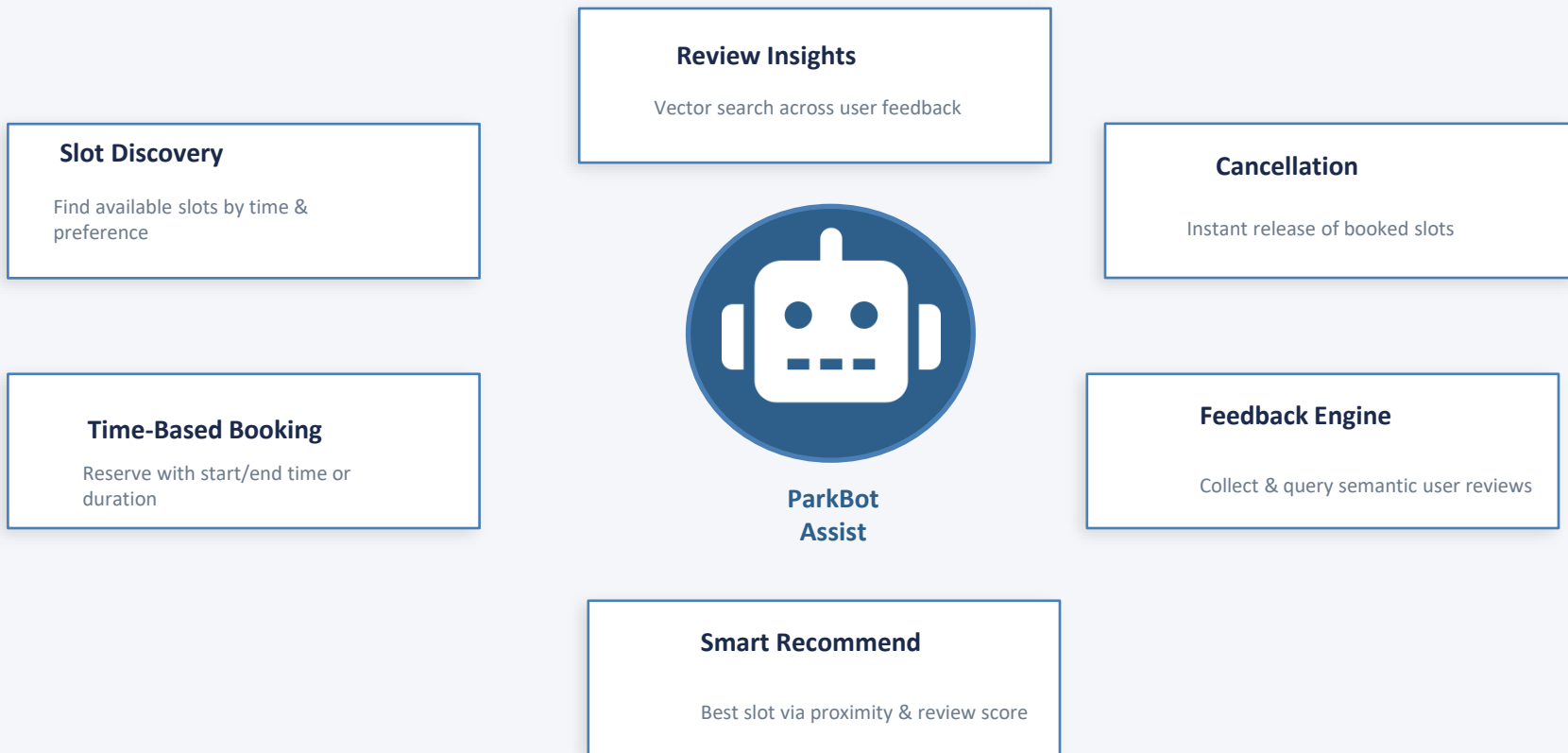
Existing systems lack time-slot reservations, causing double occupancy and no-show inefficiencies.



Lost Feedback Loop

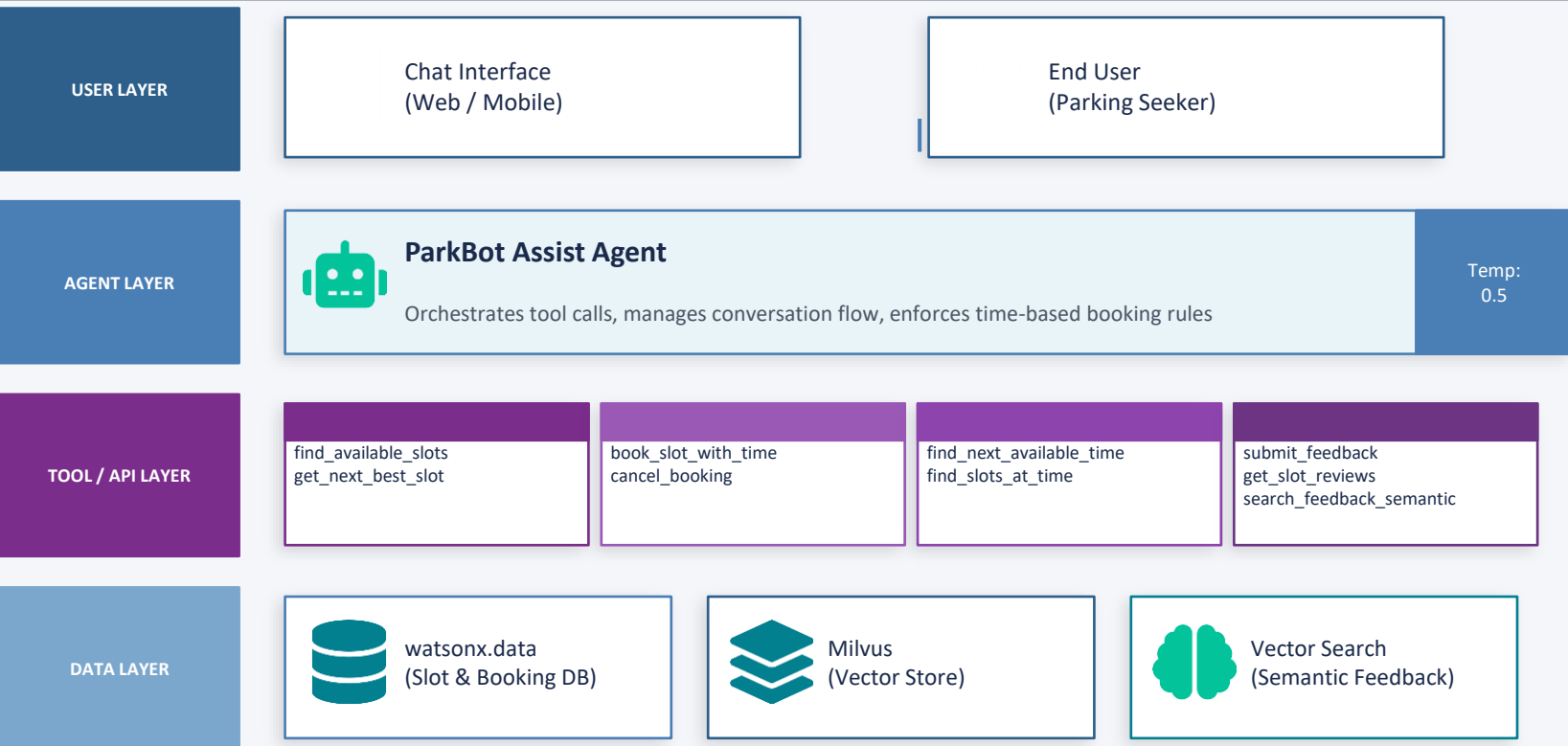
Facility operators have no mechanism to capture slot-level user reviews for service improvement.

ParkBot Assist - An AI Agent for Smart Parking



SOLUTION ARCHITECTURE

How ParkBot Assist is built and deployed



What ParkBot Assist can do for you



Smart Slot Discovery

Real-time availability lookup
Filter by level, zone, proximity
Time-aware slot search



Time-Based Reservations

Book with start + end time
'Book now for 2 hours' supported
Future bookings handled



AI Recommendations

Best slot via proximity score
Review-informed suggestions
Auto-selects when asked



Semantic Feedback Search

Vector search over reviews
'Well-lit slots' type queries
Milvus-powered similarity



Feedback Collection

Post-booking prompts
Star rating + comments
Powers future recommendations

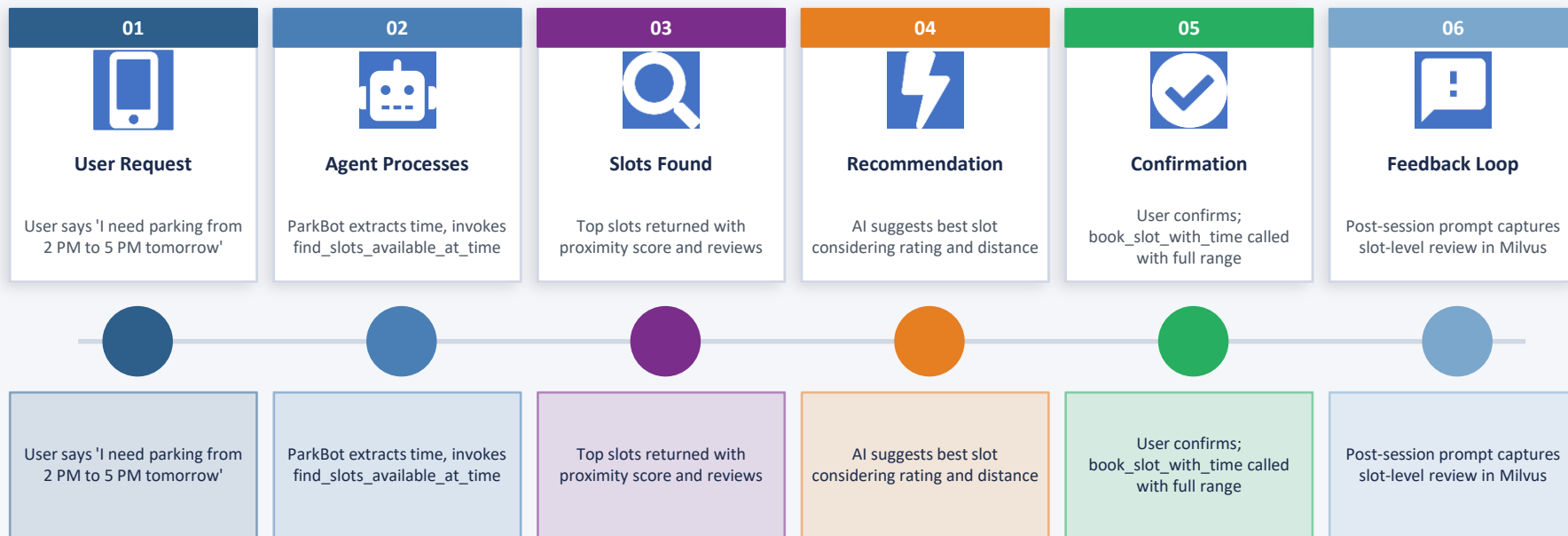


Full Booking Lifecycle

Reserve, modify, cancel
Slot status checks
Next-available time lookup

USER JOURNEY

From intent to parked — in seconds



CRITICAL RULE:

Every booking requires a time range. ParkBot NEVER confirms without a start time and end time (or duration). 'Book now for 2 hours' → start = current time, end = current + 120 mins.

Real conversations ParkBot Assist handles

S1 Instant Booking

User: *"I need parking now for 2 hours"*

ParkBot:

Recommends Slot B12, books from current time to +2 hrs, confirms location near elevator

S2 Future Reservation

User: *"I need parking tomorrow, 2 PM to 5 PM"*

ParkBot:

Searches time-specific slots, suggests Slot A05 with best reviews, books the 3-hour window

S3 Availability Check

User: *"When will slot D1 be free?"*

ParkBot:

Calls find_next_available_time, replies 'Slot A12 free after 4:30 PM', offers to book

S4 Cancel and Feedback

User: *"Cancel my booking for B2"*

ParkBot:

Releases slot instantly, asks for feedback. Stores review in Milvus for future semantic search

TECHNOLOGY STACK

Built on enterprise-grade IBM and open-source AI infrastructure



watsonx.data

Lakehouse for real-time slot availability, booking records, and occupancy state management.



Milvus

Open-source vector database stores user feedback as embeddings. Powers semantic similarity search queries.



AI Agent Framework

Native agent spec (v1) with 12 registered tools. Handles multi-turn dialog, tool selection, and time math.



IBM Cloud Deploy

Fully containerised deployment on IBM Cloud with scalable compute for peak parking demand periods.

Measurable outcomes for operators and users

70%

Reduction in
search time

24/7

AI-powered
assistance

12

Intelligent
tools available

0

Double bookings
with time locking

For End Users:

No more circling. Instant slot recommendations with proximity and review scores. Book in a natural conversation - 'parking near the elevator for 2 hours'.

For MLCP Operators:

Real-time occupancy data, time-slot level utilization analytics, and semantic feedback to identify problem slots and improve facility design.

For the Business:

Reduced staff load for booking queries, lower no-show rates with time-locked reservations, and a feedback engine that continuously improves NPS.